Experiment 16

Time Series Analysis of Website Traffic.

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

data = pd.read_csv("website_traffic.csv", parse_dates=["Date"], index_col="Date")

decomposition = seasonal_decompose(data["Page_Views"], model="additive")
decomposition.plot()
plt.show()

data["Page_Views_MA"] = data["Page_Views"].rolling(window=30).mean()
data[["Page_Views", "Page_Views_MA"]].plot()
plt.title("Page Views with Moving Average")
plt.show()

data["Page_Views_Z"] = (data["Page_Views"] - data["Page_Views"].mean()) / data["Page_Views"].std()
anomalies = data[data["Page_Views_Z"].abs() > 3]
plt.plot(data.index, data["Page_Views"], label="Page Views")
plt.scatter(anomalies.index, anomalies["Page_Views"], color="red", label="Anomalies")
plt.legend()
plt.show()
```

Output

